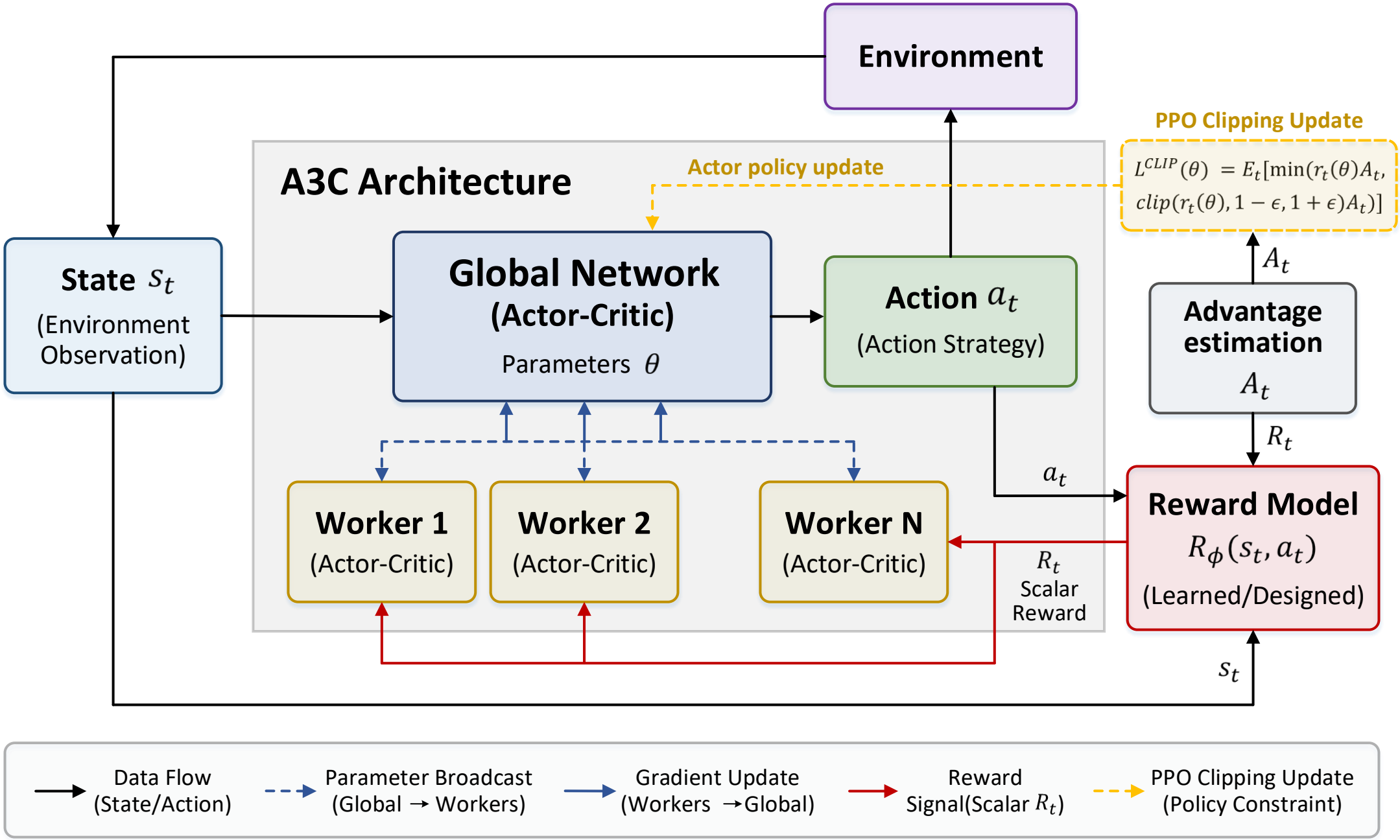

Environment
PPO Clipping Update
$L^{CLIP}(\theta) = E_t[\min(r_t(\theta)A_t, clip(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)]$
A3C Architecture
Actor policy update
State s_t
(Environment Observation)
Global Network
(Actor-Critic)
Parameters θ
Action a_t
(Action Strategy)
A_t
Advantage estimation
A_t
R_t
a_t
Worker 1
(Actor-Critic)
Worker 2
(Actor-Critic)
Worker N
(Actor-Critic)
R_t
Scalar Reward
Reward Model
$R_\phi(s_t, a_t)$
(Learned/Designed)
s_t
Data Flow (State/Action)
Parameter Broadcast (Global → Workers)
Gradient Update (Workers →Global)
Reward Signal(Scalar R_t)
PPO Clipping Update (Policy Constraint)